A fixed certificate for reciprocal Alice angles

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The source ratio can become large while the corresponding state quadratic remains positive. Two source-dependent rescalings reveal why: the remaining coupling vector has norm at most one. We use this contraction to complete the leading quadratic form, then control the three corrections in the full parent. The resulting testing functions are independent of the source.

The source and the polynomial to be certified. The source variables obey $x, \sigma, z > 0$, $0 < b, h < 1$, $r = bz < 1$. Set

$$Q = 1 - b, \quad \theta = 1 - h, \quad e = xh/\sigma, \quad E_s = 1 - e, \quad K_s = 1 + x, \quad R_s = 1 + xr,$$
$$M = R_s - xz\theta, \quad J = 1 + xh - \sigma(1 - r),$$
$$P_s = \sigma + K_s + \sigma z(1 + bx + 2xh) + \sigma b x^2 h z^2 + x^2 h\theta z,$$
$$\pi = \frac{z(1 + xh)E_s + (1 - r)(1 + xhz)}{2P_s}, \qquad \ell = \frac{(1 + \sigma r)\{eR_s(1 + \sigma z) + ME_s\}}{2P_s},$$
$$\kappa_s = \frac{K_s(1 + \sigma r)(1 + \sigma z)}{2MP_s}.$$

The strict gaps are $E_s, M, J > 0$. The seven source coefficients, in the normalization $d = 1$, are

$$m = \kappa_s x^2 z \frac{h}{\sigma}, \qquad C = \frac{K_s z(1 + \sigma r)E_s}{2(1 - r)P_s}, \qquad A = \frac{\pi}{bz}, \qquad \alpha = \frac{\ell}{Qz},$$
$$D = \frac{K_s(1 + \sigma z)(1 + \sigma r) - JP_s}{2JP_s}, \qquad \beta = \frac{K_s(1 + \sigma r)(1 + xhz) - \theta P_s}{2\theta P_s}.$$

In particular $m > 0$; the hypotheses also require $D, \beta \geq 0$.

For $D_p = 1 + p^2$, $D_w = 1 + w^2$, the complete inverse-Alice parent is

$$\begin{aligned}\widehat{F} = {} & 4m(pW - pwX + Y - wZ)^2 + 4CD_w(W - pY)^2 + 4AD_pD_wY^2 \\ & + (4D + 2)(wW + X - pwY - pZ)^2 + (4\alpha + 2)D_p(Y - wZ)^2 \\ & + (4\beta + 2)D_w(pX + Z)^2 - 2D_pD_wZ^2.\end{aligned} \tag{1}$$

It equals $p^2 F_w(1/p, w)$ for $p \neq 0$. For unit states the physical slack is $\widehat{F}/(D_pD_w)$. The pure testing factor $\kappa_t = (D_pD_w)^{-1/2}$ therefore restores the physical normalization and remains regular at $p = 0$.

Theorem 1. *Fix a real constant $P_* \geq 1$ and set*

$$\eta = 2^{-30}P_*^{-2}, \qquad \delta = 2^{-12}P_*^{-1}, \qquad f = 1 - 25\eta.$$

Use the native source just defined, normalized by $d = 1$. Suppose

$$0 < x, \sigma \leq \frac{\eta}{1 + z}, \quad z > 0, \quad 0 < b, h < 1, \quad r = bz < 1, \quad E_s, M, J > 0, \quad D, \beta \geq 0. \tag{2}$$

There is no upper bound on $H = h/\sigma$. For the independent testing coordinates

$$p = 1/u, \qquad 0 \le p \le P_*, \quad 0 \le w \le \delta,$$

the complete inverse-Alice parent has a separated square identity in one fixed finite family of testing functions. This family depends on P_* and δ, and is independent of z or any upper bound for z. The construction uses 51 testing functions. The identity includes every state vector and the literal testing face $p = 0$. Neither D nor β is inverted, so their permitted zero faces are included.

The testing statement applies to the projectively normalized parent $p^2F(1/p, w)$, continued polynomially at $p = 0$. Negative angle signs are handled by the fixed gauges $(u, Y, Z) \mapsto (-u, -Y, -Z)$ and $(w, X, Z) \mapsto (-w, -X, -Z)$. Complex states are separated into real and imaginary parts. No restriction on a ratio of state coordinates is imposed.

Source estimates. Define

$$H_0 = 1 + zQ, \qquad B_s = P_s + xz(J - \sigma),$$
$$\mathcal{H} = R_sH_0 + z\{(1 - r)xh(2 + xr) - e[2 + x(1 + h + r\theta)] + K_se^2\}.$$

The exact native identities are

$$2\theta P_s\beta = hB_s - \sigma H_0, \qquad 2P_s(JD + x\theta\beta) = \sigma^2\mathcal{H}. \tag{3}$$

Define the leading weights

$$C_0 = \frac{z}{2(1 - r)}, \qquad A_0 = \frac{H_0}{2bz}, \qquad \alpha_0 = \frac{1}{2Qz}. \tag{4}$$

The native coefficients give the exact ratios

$$\frac{C}{C_0} = \frac{K_s(1 + \sigma r)(1 - e)}{P_s}, \tag{5}$$
$$\frac{A}{A_0} = \frac{z(1 + xh)(1 - e) + (1 - r)(1 + xhz)}{P_sH_0}, \tag{6}$$
$$\frac{\alpha}{\alpha_0} = \frac{(1 + \sigma r)\{eR_s(1 + \sigma z) + M(1 - e)\}}{P_s}. \tag{7}$$

To control all three ratios together, bound the weighted source openings. Put $T = 1 + z$. From (2), $Tx, T\sigma \le \eta$. Using $bz < 1$ in the σbx^2hz^2 term shows

$$1 \le P_s \le 1 + 4\eta, \quad |M - 1| \le \eta, \quad J - \sigma > 0, \quad 1 \le B_s \le 1 + 6\eta.$$

Initially $e < 1$. Dropping the negative term in $\mathcal{H}$ yields $\mathcal{H} \le 4T$. From (3) one obtains the exact identity

$$eB_s = xH_0 + \sigma\mathcal{H} - 2P_sJD/\sigma,$$

so $e \le Tx + 4T\sigma \le 5\eta$, uniformly in H. The first ratio lies between $(1 - 5\eta)/(1 + 4\eta) \ge 1 - 9\eta$ and $(1+\eta)^2 \le 1+3\eta$. The numerator of the second ratio lies between $H_0(1-e)$ and $H_0+2xhz \le H_0(1 + 2\eta)$, since $H_0 \ge z$. It has the same lower bound $1 - 9\eta$ and upper bound $1 + 2\eta$. For the third ratio the expression in braces is a convex combination of M and $R_s(1 + \sigma z)$; they lie between $1 - \eta$ and $1 + 3\eta$. Consequently

$$|C/C_0 - 1| \le 10\eta, \quad |A/A_0 - 1| \le 10\eta, \quad |\alpha/\alpha_0 - 1| \le 5\eta. \tag{8}$$

Thus $C - fC_0$, $A - fA_0$, and $\alpha - f\alpha_0$ are nonnegative. The same fraction f of each leading weight can therefore be retained.

The exact leading completion. Set $q = \sqrt{z} > 0$ and introduce the full-state forms

$$E = qW, \quad F = Y/q, \quad G = wZ/q, \quad R = X + wW, \quad S = pX.$$

The complete core is

$$K = R^2 + (1+w^2)S^2 + z(G-F)^2 + p^2(1+w^2)zF^2 - 2pE(G+w^2F) + 2qwS(G-F). \tag{9}$$

Write

$$K_0 = R^2 + S^2 + z(G-F)^2 + p^2zF^2 - 2pEG.$$

Taking the three weights (4) at $w = 0$ gives

$$Q_0 = 2K_0 + \frac{4C_0}{z}(E - pzF)^2 + 4A_0z(1+p^2)F^2 + 4\alpha_0z(1+p^2)(F-G)^2.$$

Define

$$J_0 = F - bG, \quad B = \frac{H_0}{bQ}, \quad J_* = \sqrt{B}\,J_0, \quad G_* = \sqrt{H_0}\,G,$$
$$a = \frac{z}{\sqrt{B}}, \qquad c = \frac{1}{\sqrt{H_0}}, \qquad L = aJ_* + cG_* = zJ_0 + G.$$

Direct completion of E gives the exact identity

$$Q_0 = 2R^2 + 2S^2 + \frac{2}{1-r}(E - pL)^2 + 2(1+p^2)J_*^2 + 2G_*^2. \tag{10}$$

Indeed the p^2 remainder after completing E is $(z+1/Q)(F-bG)^2/b$, and the constant remainder diagonalizes in (J_0, G). The bounded source coefficients satisfy

$$a^2 + c^2 = \frac{bQz^2+1}{H_0} \le 1, \qquad H_0 - 1 - bQz^2 = zQ(1-r) \ge 0. \tag{11}$$

This is the reason that a large value of z causes no loss after the source-only normalization.

Put $t = p/P_* \in [0,1]$, $\bar{E} = E/P_*$, and $y = (\bar{E}, J_*, G_*, R, S)$. Let

$$B_0 = 2R^2 + 2S^2 + 2(\bar{E} - tL)^2 + 2(1+t^2)J_*^2 + 2G_*^2.$$

The leading excess is exactly

$$Q_0 - B_0 = 2\left(\frac{P_*^2}{1-r} - 1\right)(\bar{E} - tL)^2 + 2(P_*^2 - 1)t^2J_*^2. \tag{12}$$

It is nonnegative. More specifically, the following square identity proves $B_0 \succeq I_5/2$ on the independent forms y:

$$\begin{aligned} B_0 - \tfrac{1}{2}\|y\|^2 = {} & \tfrac{3}{2}(R^2+S^2) + \tfrac{3}{2}(\bar{E} - \tfrac{4}{3}tL)^2 + \tfrac{3}{2}J_*^2 + \tfrac{5}{6}G_*^2 \\ & + \tfrac{2}{3}(1-t^2)G_*^2 + \tfrac{4}{3}t^2J_*^2 \\ & + \tfrac{2}{3}t^2\{(1-a^2-c^2)(J_*^2+G_*^2) + (cJ_* - aG_*)^2\}. \end{aligned} \tag{13}$$

Every coefficient or radical in this identity belongs entirely to the source field or entirely to the testing field.

The full retained-square identity. Let $\widehat{F} = p^2 F(1/p, w)$ be the projectively normalized parent. Besides the core $2K$, its three selected events are

$$\frac{4C}{z}(1+w^2)(E-pzF)^2, \qquad 4Az(1+p^2)(1+w^2)F^2, \qquad 4\alpha z(1+p^2)(F-G)^2.$$

All remaining events are retained literally:

$$\begin{aligned}\mathcal{X} = {} & 4m(pW - pwX + Y - wZ)^2 + 4D(wW + X - pwY - pZ)^2 \\ & + 4\beta(1+w^2)(pX+Z)^2 \\ & + \frac{4(C - fC_0)}{z}(1+w^2)(E-pzF)^2 \\ & + 4(A - fA_0)z(1+p^2)(1+w^2)F^2 \\ & + 4(\alpha - f\alpha_0)z(1+p^2)(F-G)^2.\end{aligned}$$

They have nonnegative source weights. Define the further retained sum

$$\begin{aligned}\mathcal{P} = {} & 2(1-f)\{R^2 + S^2 + z(G-F)^2 + p^2 z F^2\} \\ & + 2w^2 S^2 + 2p^2 w^2 z F^2 \\ & + \frac{4fC_0}{z} w^2 (E - pzF)^2 + 4fA_0 z(1+p^2) w^2 F^2.\end{aligned}$$

Then the complete parent, with no omitted state terms, is

$$\widehat{F} = fQ_0 + \mathcal{E} + \mathcal{P} + \mathcal{X}, \tag{14}$$

where the only correction is

$$\mathcal{E} = -4(1-f)pEG - 4pw^2 EJ_0 - 4pbw^2 EG - 4qwSJ_0 + 4qQwSG. \tag{15}$$

This follows from (9); its difference from K_0 includes the positive term $p^2 w^2 z F^2$, which has been retained in $\mathcal{P}$.

Explicit source Gram assignments. In the ordering $y = (\bar{E}, J_*, G_*, R, S)$ write

$$\mathcal{E} = y^{\mathsf{T}}(tC_1 + tw^2 C_2 + wC_3)y.$$

Only the following upper-triangular entries are nonzero; the lower triangles are obtained by symmetry:

$$\begin{aligned}&(C_1)_{13} = -2(1-f)P_*^2/\sqrt{H_0}, \\ &(C_2)_{12} = -2P_*^2/\sqrt{B}, \qquad (C_2)_{13} = -2bP_*^2/\sqrt{H_0}, \\ &(C_3)_{25} = -2q/\sqrt{B}, \qquad (C_3)_{35} = 2qQ/\sqrt{H_0}.\end{aligned}$$

The row norms are at most $50\eta P_*^2, 4P_*^2, 4$, respectively. For the last bound use

$$q^2/B = rQ/H_0 \le 1, \qquad q^2 Q^2/H_0 = zQ^2/H_0 \le 1.$$

The first two use $H_0 \ge 1$ and $B \ge 1$. Set

$$\begin{gathered}T_1 = C_1, \quad T_2 = \delta^2 C_2, \quad T_3 = \delta C_3, \\ M_1 = 50\eta P_*^2, \quad M_2 = 4P_*^2\delta^2, \quad M_3 = 4\delta, \quad \rho = M_1 + M_2 + M_3.\end{gathered}$$

Then

$$\rho \le 50 \cdot 2^{-30} + 4 \cdot 2^{-24} + 4 \cdot 2^{-12} < 1/512, \qquad f/2 - \rho > 1/4.$$

All six source matrices $(M_iI_5 \pm T_i)/2$ are positive semidefinite. With pure testing monomials

$$m_1 = t, \quad m_2 = t(w/\delta)^2, \quad m_3 = w/\delta,$$

one has the exact identity

$$\rho\|y\|^2 + \mathcal{E} = \sum_{i=1}^{3} \left\{ (1+m_i) y^{\mathsf{T}} \frac{M_iI_5 + T_i}{2} y + (1-m_i) y^{\mathsf{T}} \frac{M_iI_5 - T_i}{2} y \right\}. \tag{16}$$

Also

$$fQ_0 - \rho\|y\|^2 = f(Q_0 - B_0) + f(B_0 - \tfrac{1}{2}\|y\|^2) + (f/2 - \rho)\|y\|^2.$$

Equations (12), (13), and (16), together with $\mathcal{P} + \mathcal{X}$, specify all positive source Gram blocks of the complete parent.

An explicit list of 51 testing functions. Fix $P_* \geq 1$, put $t = p/P_*$, and write

$$\phi = (X, Y, W, wW, wZ, pX)^T, \qquad y_1 = t, \quad y_2 = t(w/\delta)^2, \quad y_3 = w/\delta.$$

Take the 42 labelled functions consisting of every entry of ϕ multiplied by one of

$$1, \quad \sqrt{1+y_1}, \sqrt{1-y_1}, \quad \sqrt{1+y_2}, \sqrt{1-y_2}, \quad \sqrt{1+y_3}, \sqrt{1-y_3},$$

and the nine further functions

$$pwZ,\ pY,\ pwY,\ wY,\ pwX,\ pW,\ pZ,\ Z, \quad wZ\sqrt{1-t^2}.$$

All radicands are nonnegative on $0 \leq p \leq P_*$, $0 \leq w \leq \delta$. This list is independent of all source variables. For the physical slack multiply its entries by κ_t. For signed angles compose the list with the fixed angle-sign and state-sign gauges. No state-coordinate inverse or sign-dependent source factor occurs.

Polynomial weighted blocks. Here is the complete smaller weighted representation. A row (v, g, H) means $g\, v^T H v$. Write $[v] = vv^T$, and put

$$t_B = \sqrt{B}, \quad t_H = \sqrt{H_0}, \quad \ell_J = (t_B/q, -bt_B/q)^T, \quad \ell_G = (0, t_H/q)^T,$$
$$h_s = \frac{1}{P_*}(q, -sz/q, -s(1-r)/q)^T \quad (s = 1, 4/3),$$
$$\nu = 1 - a^2 - c^2 = \frac{Qz(1-r)}{H_0}, \qquad \psi = (Y, wZ)^T.$$

Define source matrices

$$G_{RS} = (2 - f/2) I_2,$$
$$G_{EL} = \tfrac{3}{2} f [h_{4/3}] + 2f \left(\frac{P_*^2}{1-r} - 1 \right) [h_1],$$
$$G_{JG} = \tfrac{3}{2} f [\ell_J] + \tfrac{5}{6} f [\ell_G] + 2(1-f)[(1,-1)^T],$$
$$G_{pJG} = f \left(2 - \frac{2}{3P_*^2} \right) [\ell_J] + \frac{2f}{3P_*^2} \{ \nu([\ell_J] + [\ell_G]) + [c\ell_J - a\ell_G] \} + 2(1-f)[(1,0)^T].$$

Their coefficients are nonnegative. In the order $(R, S, J_*, \bar{E}, G_*) = \mathsf{L}\phi$, the source matrix is

$$\mathsf{L} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & t_B/q & 0 & 0 & -bt_B/q & 0 \\ 0 & 0 & q/P_* & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & t_H/q & 0 \end{pmatrix}.$$

Let Π be the permutation with $\Pi y = (R, S, J_*, \bar{E}, G_*)^T$, and define $\widetilde{T}_i = \Pi T_i \Pi^{\mathsf{T}}$ from the three matrices in the theorem. Let M_i be their stated row-sum bounds. Thus the seven main rows are

$$\left(\phi, 1, \mathsf{L}^T(f/2-\rho)I_5\mathsf{L}\right), \qquad \left(\phi, 1 \pm y_i, \mathsf{L}^T \frac{M_i I_5 \pm \widetilde{T}_i}{2}\mathsf{L}\right), \quad i = 1, 2, 3.$$

The other twelve rows are

v	g	H
$(X + wW, pX)^T$	1	G_{RS}
$(W, pY, pwZ)^T$	1	G_{EL}
ψ	1	G_{JG}
wZ	$1 - t^2$	$2ft_H^2/(3q^2)$
ψ	p^2	G_{pJG}
$(pwX, pwY)^T$	1	$2I_2$
$(W - pY, w(W - pY))^T$	1	$4\,\mathrm{diag}(C - fC_0, C)$
$(Y, wY)^T$	D_p	$4\,\mathrm{diag}(A - fA_0, A)$
$pW - pwX + Y - wZ$	1	$4m$
$wW + X - pwY - pZ$	1	$4D$
$pX + Z$	D_w	4β
$Y - wZ$	D_p	$4(\alpha - f\alpha_0)$

Their sum is exactly (1). There are 19 blocks, of total order

$$7 \cdot 6 + (2 + 3 + 2 + 1 + 2) + 2 + 2 + 2 + 4 = 62,$$

with maximum order six. All matrix entries are source-only. They are explicit sums or congruences of nonnegative source rank-one matrices, diagonal matrices, or the proved row-sum matrices. The testing scalar multipliers are nonnegative polynomials.

To recover the stated scalar list, factor each source Gram in the source field and split $D_p = 1 + p^2$, $D_w = 1 + w^2$, and p^2 into polynomial testing squares. Every resulting ungated root is a source-linear combination of the fourteen displayed monomials; the only remaining radicals are the six main gate roots and $\sqrt{1 - t^2}$ multiplying wZ. Thus no additional testing function is introduced by any of the retained full events.

Corollary 1 (All Alice angles in the bounded chart)**.** *Choose $P_* = 2^{24}$. On the native source domain*

$$0 < z \le 1, \qquad 0 < x, \sigma \le \frac{2^{-78}}{1 + z},$$

the union of this construction and the inner-angle certificate[1] *covers every projective Alice angle and $|w| \le 2^{-36}$, for every state and arbitrary H.*

Proof. The present theorem covers $|u| \ge 2^{-24}$, including infinity. For $|u| \le 2^{-24}$, the positive-λ inner testing condition $\sqrt{|u|}, |w| \le 2^{-12}$ holds. The nonpositive branch has the weaker requirement $|u|, |w| \le 2^{-12}$. The present source box is contained in both inner source boxes. The cases meet at $|u| = 2^{-24}$ and include $u = 0$ literally. □

The rational affine pencil. The weighted vectors in the table determine a finite semidefinite representation without retaining the nonlinear source formulas as constraints. Let $c = (d, m, C, A, D, \alpha, \beta)$, and let G_b be unrestricted positive semidefinite matrices of the displayed orders. Define the certificate cone by

$$\widehat{F}_c = \sum_{b=1}^{19} g_b v_b^{\mathsf{T}} G_b v_b, \qquad G_b \succeq 0.$$

[1] A fixed finite C1 testing family and an evaluated rational local pencil, Theorem 1.

The identity is an equality of polynomials in p, w, W, X, Y, Z. Every v_b and g_b is fixed once P_* is chosen. Their rational coefficients make coefficient comparison a rational linear system

$$\mathcal{A}\,\gamma = \mathcal{B}\,c,$$

where γ lists the 176 upper-triangular Gram entries. For $P_* = 2^{24}$ there are 80 coefficient equations of rank 79. Eliminating the pivot entries leaves 97 free entries y and gives the rational pencil $L_{62}(c, y)$ of total order 62. The accompanying `SOURCE_PENCIL.coo.json` lists its 338 nonzero affine coefficients.

The two directions of this representation follow from the same elimination. A feasible pencil reconstructs positive semidefinite G_b satisfying every polynomial coefficient equation. Conversely, for any certificate, take its 97 free Gram entries as y; the pivot equations reconstruct the remaining entries of those same matrices. Thus the pencil represents this weighted certificate cone exactly.

Sharing c with the inner-angle pencil gives the intersection

$$L_{731}(c, y, z) = \operatorname{diag}\big(L_{669}(c, y), L_{62}(c, z)\big).$$

It has 139 blocks and 2791 auxiliary coordinates. Its source admission and testing coverage are precisely those of Corollary 1. The file `ALL_ALICE_ORDER731.coo.json` gives all its entries.

Remaining domains. For $0 < z \le R$ one sufficient admission box is $x, \sigma \le 2^{-30} P_*^{-2}/(1 + R)$; the testing family is unchanged as R varies. Every germ with x, σ infinitesimal and z bounded satisfies (2). Thus this is one testing family for the inverse-Alice part of all those bounded-z germs, without choosing a real upper bound for z first. It also covers unbounded z whenever $(1 + z)x, (1 + z)\sigma \le \eta$. It does not cover a native source with larger weighted openings, nor reciprocal Bob angles $|w| > \delta$. For $z > 1$, the existing inner extension has an R-dependent testing interval; the present outer identity alone does not remove that separate inner issue. Literal zero openings excluded by the native formulas, such as $z = 0$, are handled in their other source charts and are not obtained by division here. No full-Q_{222} adequacy statement follows.